

Zero-Shot Real-Robot Transfer: Wrench-Nut Insertion

Task: insert a grasped wrench onto a nut threaded on an upright bolt. Harder than peg-in-hole: **the nut rotates freely under contact**, so the target pose shifts mid-attempt and the policy must keep realigning.

7-DoF Kuka iiwa14 + Robotiq 3-Finger gripper. Trained **only in simulation** (IsaacGym), deployed **zero-shot**; FoundationPose 6D tracking from one RGB-D camera. Success: within **5 mm** of the nut center in **128 steps**. Both methods: same observations, actions, controller, 500K updates.

Ours (full method)

[video: ours]

74% success (in-distribution, 30 trials)
lower position (**9.9 vs. 11.1 mm**) and rotation error on successful insertions

⇒ **+13 points zero-shot, with no real-world fine-tuning, on a non-humanoid platform**

Ordering holds on **unseen** nut sizes: 80 vs. 73% (size 3) and 33 vs. 20% on the hardest size 1; Wilson CIs overlap at these trial counts, but ours leads on every size.

TD-M(PC)² (SOTA baseline)

[video: baseline]

61% success (same 30 trials, same stack)
identical training and deployment; **[failure mode from video, e.g., stalls mis-aligned until the time limit]**